

LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

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Abstract

Multicenter medical models are typically trained asynchronously across hospitals with heterogeneous acquisition devices, patient populations, and diagnostic-label coverage. We study single-shot asynchronous medical model merging, where each client uploads its locally trained model only once and the server constructs a global model without accessing raw medical images or conducting federated optimization rounds. We identify aggregation-induced prediction collapse as the central failure mode of general-purpose post-hoc merging methods. Furthermore, we show that long-tailed medical priors can make majority-class collapse deceptively competitive in terms of raw accuracy, thereby masking this failure. To address this issue, we propose LAMP-Merge, a long-tail-aware medical prototype-merging framework that reconstructs a diagnostic prototype head from client-side class statistics and calibrates long-tail prevalence using a bounded prior bias. Across five medical datasets, five model families, three client-population sizes, and three levels of non-IID skew, LAMP-Merge matches or surpasses the strongest non-LAMP baseline in 68 of 75 client-averaged settings, while improving balanced accuracy, macro-F1, and prediction-collapse diagnostic metrics.

Index Terms: medical model merging, asynchronous fusion, predictive collapse, long-tail calibration, privacy-preserving learning

I Introduction

Medical artificial intelligence is transitioning from single-center model development toward multicenter collaborative deployment. The reliability of clinical diagnostic models typically depends on experience spanning hospitals, imaging devices, and patient populations, as no single institution can readily cover the full disease spectrum, rare cases, and diverse imaging conditions simultaneously. However, medical data are jointly constrained by ethical approval requirements, privacy regulations, device heterogeneity, and institutional workflows, making centralization of raw images on a common server difficult. A more practical collaboration paradigm is for hospitals to train models locally and, upon completion, participate in global model construction through limited information sharing.

This practical setting motivates a need for single-shot asynchronous model merging. Hospitals may not be online simultaneously or be able to repeatedly re-train after each global-model update; some centers may finish training earlier, whereas others may upload their models only within fixed time windows. Meanwhile, at deployment time, the server must produce a globally usable diagnostic model rather than a collection of independent local models with limited coverage. Accordingly, we study a post-hoc merging setting in which each client uploads exactly once and the server synthesizes a shared checkpoint without compromising patient privacy, i.e., without accessing clients’ raw medical images. Medical imaging makes this problem particularly challenging. Inter-hospital heterogeneity arises not only from image-style and device-domain shifts, but also from differences in diagnostic-class availability: one center may accumulate abundant samples for common diseases or routine examinations while having virtually no cases of certain rare diagnoses, whereas another may exhibit the opposite case composition. The resulting local models are therefore not independent replicas of a complete global task; rather, they are partial experts containing localized diagnostic evidence. More importantly, medical class distributions are typically long-tailed: majority diagnoses may account for a substantial fraction of samples, whereas minority diagnoses remain clinically meaningful. An effective merging method must both preserve class-specific evidence distributed across clients and avoid misinterpreting genuine long-tail prevalence as irrelevant noise.

We propose LAMP-Merge, a long-tail-aware medical prototype-merging framework. Its central idea is to recast medical model merging from a problem of global parameter averaging into one of class-level diagnostic-evidence reconstruction. Using a shared reference feature extractor, each client computes an aggregated prototype and sample count for every diagnostic class locally, and uploads only these class-level statistics together with an interface to its local checkpoint. The server then constructs a global diagnostic prototype classifier in the shared reference space and introduces bounded prior calibration when class prevalence exhibits a pronounced long-tailed structure. The entire

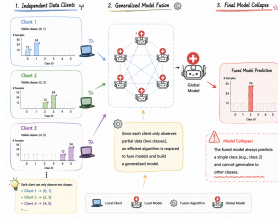


Figure 1: In multicenter healthcare settings, individual clients typically observe only a subset of the global diagnostic space. Conventional parameter-level model-merging methods can therefore cause the merged global model’s predictions to concentrate on a small number of dominant classes.

procedure requires only a single upload and does not require the server to access raw images, per-sample features, per-sample logits, or per-sample predictions.

Our main contributions are as follows:

- We introduce and formalize the problem of single-shot asynchronous medical image model merging, in which clients train independently on local data and the server must construct a global model using information from a single upload.
- We characterize the key structures that distinguish this setting from general model-merging tasks, namely heterogeneous diagnostic-class availability across medical institutions and long-tailed disease prevalence.
- We propose LAMP-Merge, which reconstructs a global diagnostic classifier from client-side class prototypes and prevalence statistics, mitigating post-merging prediction collapse while preserving the privacy of raw medical images.

We conduct extensive experiments on multiple medical imaging datasets. The results show that LAMP-Merge consistently and substantially outperforms all competing methods across the considered data distributions. Moreover, our merging approach requires neither an auxiliary dataset nor additional computational resources during training.

II Related Works

A. Generic Post-hoc Model Merging

General post-hoc model merging aims to integrate already trained model checkpoints, thereby avoiding the costs of joint training or retraining. Existing methods can be categorized by their merging variables. The first category directly interpolates in weight space, including weight averaging and Model Soups; these methods can merge effectively when models remain in compatible parameter basins. The second category operates in parameter-delta space. For example, Task Arithmetic represents a fine-tuned model as a parameter displacement relative to a shared initialization and linearly combines such displacements. The third cate-

gory seeks to remove parameter interference before averaging. Git Re-Basin reduces permutation mismatch through neuron-permutation alignment, whereas TIES-Merging and the DARE family mitigate update conflicts by pruning small updates, applying sign voting, or randomly dropping and rescaling task-vector components. The fourth category further incorporates assumptions based on geometry, curvature, or robust statistics, including Fisher merging, RegMean, Model Stock, Breadcrumbs, Iso-C, FreeMerge, and Robust-Merge.

These methods constitute important baselines in this work, but their merging variables do not model medical diagnostic classes explicitly. They operate on global parameters, task vectors, curvature estimates, permutation structures, or update signs, rather than addressing whether client i possesses reliable evidence for diagnostic class c . In multicenter medical imaging tasks, local client datasets commonly exhibit both missing classes and long-tailed distributions. Consequently, reliable class-specific directions from clients that have observed a particular class are mixed with absent or conflicting directions from clients that have not. The merged model may therefore preserve majority-class directions while suppressing rare or locally absent classes, yielding a globally collapsed model with a highly concentrated predictive distribution.

B. Medical Model Merging

Medical fusion methods typically exploit structures specific to clinical data. For example, mTAN employs continuous-time attention for irregularly sampled time series, Raindrop uses graph structures to model relationships among physiological sensors, and MedFuse integrates chest X-ray images with clinical time series. These studies demonstrate that medical data are not merely a straightforward application domain for general-purpose fusion algorithms; they exhibit irregular temporal structures, modality heterogeneity, clinical measurement processes, and disease-prevalence distributions.

However, these medical fusion methods typically combine patient-level input modalities or clinical representations rather than independently trained medical image-classifier checkpoints. Their training procedures often require access to patient-level samples, timestamps, multimodal records, electronic health records, or server-side validation feedback. When the server cannot access clients’ raw medical images, per-sample features, per-sample logits, or per-sample predictions, these methods cannot be directly applied to the single-shot model-merging setting studied here. Medical model-merging methods that avoid disclosing raw images also frequently rely on validation data or iterative selection procedures, which conflicts with our single-upload protocol. The recently proposed MedMerge framework in **IEEE Transactions on Radiation and Plasma Medical Sciences** is an important related work on knowledge transfer for medical vision-language models; how-

ever, it targets a different cross-modal adaptation setting rather than asynchronous checkpoint merging for the same task. In contrast, LAMP-Merge restricts medical structure to uploadable class-level statistics: clients compute diagnostic-class prototypes and prevalence counts locally, and the server constructs the model solely from these aggregated statistics.

C. Federated and Decentralized Medical Learning

Federated learning and decentralized learning are important paradigms for privacy-sensitive medical AI. FedAvg and its variants train a global model through multiple communication rounds: the server distributes a model, clients perform local optimization, and the server aggregates the uploaded updates. Swarm Learning removes the conventional central parameter server and employs decentralized coordination for multi-party training. Gossip Learning further propagates models asynchronously in a peer-to-peer manner, allowing participants to gradually benefit from collaboration through local training and model exchange.

These methods are appropriate when clients can remain online for extended periods, communicate repeatedly, and continue training from intermediate global models. However, the problem studied here is not collaborative training, but single-shot asynchronous merging after local training has been completed. In real-world medical deployments, hospitals may complete training at different times and may be unable to bear the cost of repeated model distribution, training, and upload protocols. Accordingly, in our setting, the server does not distribute a sequence of global models to clients, and clients do not continue optimization after observing the server’s merged result. Federated and decentralized learning therefore address multi-round training protocols, whereas LAMP-Merge addresses single-shot post-hoc merging. LAMP-Merge requires each client to upload its model and class-level statistics only once after local training, enabling the server to produce a unified global diagnostic model without establishing a multi-round feedback loop.

Medical image model merging requires an explicit representation of class-level diagnostic availability while preserving patient privacy without multi-round communication. LAMP-Merge is designed to address this gap: clients upload class prototypes and class-support counts in a single round, and the server reconstructs a global diagnostic prototype classifier on a per-class basis, thereby rebuilding the class-specific diagnostic head.

III Method

A. Motivation and Overall Idea

We find that, in medical settings with partial local class coverage, post-hoc parameter merging can transform multiple non-degenerate local predictors into a globally collapsed predictor. When the most prevalent diagnostic class accounts for a large proportion of the data dis-



Figure 2: Local models from multicenter hospitals typically contain only partial diagnostic knowledge, while raw medical images cannot be centrally shared. LAMP-Merge reconstructs a privacy-friendly global medical classifier through diagnostic prototype reconstruction and long-tail prevalence calibration, without accessing raw data or requiring multi-round federated optimization.

tribution, such a collapsed predictor can achieve high accuracy despite lacking genuine multiclass diagnostic capability. This explains why some baselines remain competitive in terms of raw accuracy. An effective medical merging method should avoid losing discriminative capability for rare classes while respecting true prevalence information when the task genuinely exhibits a strongly long-tailed distribution.

To address these challenges, we propose LAMP-Merge, a long-tail-aware medical model-merging framework based on class-level diagnostic prototype reconstruction. It recovers multiclass discriminative directions over the complete diagnostic space, structurally mitigating post-merging prediction collapse toward a single class or a small subset of classes. However, because removing class priors may impair overall accuracy, we further design a long-tail prevalence calibration module that estimates the global diagnostic prior from client-uploaded class-prevalence counts. Specifically, LAMP-Merge comprises two core components: (1) a mechanism for aggregating reliable class-level knowledge across multiple local medical clients and reconstructing a global diagnostic classifier resistant to prediction collapse; and (2) a strategy for preserving clinically meaningful majority-class priors while maintaining multiclass discriminative capability. This design enables more balanced and effective integration of diagnostic knowledge, allowing robust single-shot medical model merging without additional federated training or exposure of raw medical data.

B. Overview

The core framework of LAMP-Merge is illustrated in figure2. Its central idea is to recast medical model merging from global parameter averaging into the reconstruction of class-conditional diagnostic evidence, supplemented by class-prior information.

Notably, LAMP-Merge neither requires the server to access clients’ raw images nor relies on a server-side val-

idation set, per-sample features, or per-sample predictions. Clients upload only class-level statistics; thus, the framework preserves a single-shot asynchronous communication boundary, distinguishing it from federated learning procedures that require repeated model broadcasts and local retraining.

The framework comprises two key components: a class-level diagnostic prototype reconstruction module and a long-tail prevalence calibration module. The prototype reconstruction module restores multiclass discriminative directions across the complete global diagnostic space, after which the calibration module retains appropriate class-prior information.

C. Diagnostic Prototype Reconstruction Module

Assume K medical clients. Client i owns a private labeled dataset $D_i = \{(x, y)\}$ drawn from a class-skewed medical distribution over C diagnosis classes. Let $D_{i,c}$ denote the subset of client i belonging to diagnosis class c . In medical partitions, $D_{i,c}$ can be empty for many classes. Thus, a client can be reliable for a subset of diagnoses but unreliable for unseen classes. A class-agnostic parameter merge has no explicit mechanism to preserve this local reliability structure.

Therefore, LAMP-Merge first reconstructs a class-wise diagnostic head in the shared reference feature space. Let ϕ_0 be the frozen reference backbone instantiated from the shared model configuration before local training, and let $T(\cdot)$ be the evaluation transform. For each diagnosis class c , client i computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y) \in D_i} \mathbf{1}[y=c] \phi_0(T(x)), \quad (1)$$

where $n_{i,c} = |D_{i,c}|$ is the class support count. If $n_{i,c} = 0$, the client uploads zero support for that class and contributes no prototype evidence.

The server converts support counts into evidence weights:

$$e_{i,c} = (n_{i,c} + 1)^\gamma \mathbf{1}[n_{i,c} > 0], \quad (2)$$

where $\gamma = 0.45$ is the default evidence exponent. The normalized reliability of client i for class c is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^K e_{j,c}}. \quad (3)$$

The global diagnosis prototype is then

$$p_c = \sum_{i=1}^K \alpha_{i,c} \mu_{i,c}. \quad (4)$$

Finally, LAMP-Merge synthesizes a cosine prototype classifier:

$$w_c = s \frac{p_c}{\|p_c\|_2}, \quad (5)$$

where $s = 20$ is the default head scale. This module addresses a key limitation of existing methods: it preserves the classes for which each client has supporting evidence and reconstructs a global multiclass diagnostic head.

D. Long-tail Prevalence Calibration

After obtaining this newly reconstructed diagnostic head, each client additionally uploads class-prevalence counts $m_{i,c}$ to account for the long-tailed prevalence inherent in real-world medical tasks. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^K m_{i,c}}{\sum_{k=1}^C \sum_{i=1}^K m_{i,k}}. \quad (6)$$

We measure dominant-class imbalance by

$$r = C \max_c \pi_c. \quad (7)$$

The module is activated only when $r > \tau$, where $\tau = 2.5$ by default. When activated, it adds a centered log-prior bias

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_{k=1}^C \log \pi_k \right), \quad (8)$$

with default maximum calibration strength $\lambda = 5.0$. The final score is

$$\text{score}_c(x) = w_c^\top \phi_0(T(x)) + b_c. \quad (9)$$

The bias is centered, bounded, and attached to the prototype classifier; it calibrates prevalence without replacing the multi-class diagnostic directions reconstructed by the Diagnostic Prototype Reconstruction Module.

Experiments

A. Experimental Settings

1) Base Models and Merging Setup: Experiments follow a single-shot asynchronous model-merging protocol. Each client first independently trains a local model on its own medical imaging data, then computes the class-level statistics required by LAMP-Merge locally, including class-support counts, reference-space class prototypes, and class-prevalence counts. These statistics are uploaded once together with an interface to the local checkpoint. The server subsequently performs a single post-hoc merging step using only the uploaded checkpoints and class-level statistics, and evaluates the resulting merged model on a global test set.

We evaluate five model families. The standard vision backbones include ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny, and the vision-language backbone is CLIP-ViT-B/32. For each dataset-model pair, we construct non-IID client partitions with $K \in \{3, 5, 7\}$ clients and Dirichlet skew parameter $\beta \in \{0, 0.01, 0.1\}$, using seed 42. A client-average result fixes the dataset, model, and K , and then averages over the three β values.

All experiments were run on a server with two Intel(R) Xeon(R) Silver 4210 CPUs, four NVIDIA GeForce RTX 2080 Ti GPUs, and the conda environment MM with Python 3.10.20, PyTorch 2.6.0+cu118, and CUDA runtime 11.8.

Table 1: Dataset statistics used in the experiments.

Dataset	Classes	Channels	Train	Val	Test
Blood	8	3	11959	1712	3421
Derma	7	3	7007	1003	2005
Organ-C	11	1	12975	2392	8216
Organ-S	11	1	13932	2452	8827
Ultrasound	8	3	3869	549	1113
Total	–	–	49742	8108	23582

2) **Evaluation Datasets:** We evaluate the merged models on multiple representative medical image benchmarks. These datasets cover key medical imaging modalities, including blood-cell microscopy, dermoscopy, abdominal CT organ slices, and ultrasound images, and involve cell morphology recognition, skin-lesion diagnosis, organ-structure recognition, and ultrasound-structure discrimination. The datasets used in the experiments include:

- **Blood:** an eight-class blood-cell microscopy dataset for hematological morphology classification, focusing on fine-grained cellular appearance and inter-class morphological similarity.
- **Derma:** a seven-class dermoscopic skin-lesion dataset, characterized by strong long-tailed prevalence and visually subtle lesion categories.
- **Organ-C:** an 11-class abdominal CT dataset in the coronal view, evaluating organ-category recognition under grayscale cross-sectional imaging.
- **Organ-S:** an 11-class abdominal CT dataset in the sagittal view, complementing Organ-C with a different anatomical imaging plane.
- **Ultrasound:** an eight-class ultrasound image dataset, covering structure recognition under modality-specific speckle noise and low-contrast boundaries.

These benchmark datasets cover diverse imaging settings in medical image classification, providing a systematic testbed for evaluating the cross-modality robustness, class-discriminative capability, and task generalization of merged models. We conduct systematic experiments on these datasets to analyze the performance of LAMP-Merge across different medical image types and class-distribution conditions.

3) **Evaluation Metrics:** The main experiments use only Accuracy (ACC) as the model-selection and cross-method comparison metric. Given N test images, let y_n denote the ground-truth label and $\hat{y}_n$ denote the predicted label. ACC is defined as

$$\text{ACC} = \frac{1}{N} \sum_{n=1}^N \mathbf{1}[\hat{y}_n = y_n]. \quad (10)$$

4) **Hyperparameter Settings:** Unless otherwise stated, LAMP-Merge uses one fixed hyperparameter configuration for all datasets, backbones, client counts, and skew levels: evidence exponent $\gamma = 0.45$, prototype head scale $s = 20$, long-tail activation threshold $\tau = 2.5$, and prevalence calibration strength $\lambda = 5.0$.

B. Comparison With the State of the Arts

We compare LAMP-Merge with twelve post-hoc merging baselines: avg, ties, dare_linear, dare_ties, reg-mean, fisher, breadcrumbs, model_stock, from, iso_c, free_merge, and robustmerge. These baselines cover representative routes in current post-hoc model merging, including weight averaging, task-vector trimming, sign-conflict resolution, curvature or mean correction, model stock, and robust merging. The main comparison uses client-average ACC: for each fixed dataset, model, and client number, we first average over the three β skew levels and then compare merging methods. A cell is counted as successful when LAMP-Merge is greater than or equal to the strongest non-LAMP baseline in that cell.

Tables 2–6 report the full client-average results. Across the 75 client-average cells, LAMP-Merge wins or ties the strongest non-LAMP baseline in 68 cells, corresponding to 90.7%. This result indicates that the gain is not an accidental effect of one client number, one partition skew, or one model family; instead, it appears consistently across multiple medical image modalities and backbone architectures.

C. Ablation Study

Module-level ablations use exactly the same experimental setup as the main experiments. We compare three configurations: full LAMP-Merge, diagnostic prototype reconstruction only, and weight averaging augmented only with long-tail prevalence calibration. The results show that both modules provide substantial performance improvements.

1) **Effectiveness of Diagnostic Prototype Reconstruction** To further identify which information within diagnostic prototype reconstruction is necessary, we conduct controlled substitutions across the full setting. All configurations retain long-tail prevalence calibration and vary only the prototype semantics or support-count weighting used in diagnostic prototype reconstruction.

The results show that the performance of diagnostic prototype reconstruction does not arise from adding an arbitrary classifier head. Removing class-conditional information, permuting diagnostic semantics, or using local classifier heads outside the shared reference space all leads to substantial degradation. Replacing class-support counts with uniform weights, binary presence indicators, or total client sizes also reduces performance. These findings indicate that the module’s effectiveness depends jointly on two factors: class prototypes in the shared reference space provide diagnostically meaningful semantic directions, while class-level support counts provide reliability weights for each client-class pair.

2) **Effectiveness of Long-tail Prevalence Calibration** We further replace the prior term inside long-tail prevalence calibration. This group changes only the class prior used by the calibration bias, and the second column gives the corresponding formula.

Table 2: Client-average test accuracy on resnet. Each entry averages the three Dirichlet settings for a fixed client number. Best in bold and second with underline. These notes are the same to others.

Method	Blood			Derma			Organ-C			Organ-S			Ultrasound		
	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.3345	0.2467	0.2822	0.6291	0.4700	0.6244	0.2261	0.1772	0.1396	<u>0.2864</u>	0.1794	0.2475	0.2435	0.1581	0.1542
ties	0.2601	0.1872	0.1787	0.5995	0.4841	0.6484	0.2650	0.1684	<u>0.2118</u>	0.1981	0.1915	0.1874	<u>0.3477</u>	0.1989	0.1854
dare_linear	0.2832	<u>0.2856</u>	0.2392	0.6357	0.4841	0.6288	0.2018	<u>0.2588</u>	0.1650	0.2545	0.2245	<u>0.2537</u>	0.2162	0.1728	0.1608
dare_ties	0.3194	0.1935	0.1780	0.5375	0.4825	0.5558	0.1851	0.1515	0.1859	0.1790	0.1352	0.1730	0.2932	0.1635	0.1890
regmean	<u>0.3460</u>	0.2369	<u>0.3407</u>	0.5812	0.2843	0.5022	0.2405	0.1340	0.1388	0.2504	0.1623	0.1493	0.2276	0.1695	0.1318
fisher	0.3196	0.2237	0.2608	0.6707	<u>0.6702</u>	<u>0.6690</u>	<u>0.2973</u>	0.1741	0.1812	0.2010	<u>0.2956</u>	0.1963	0.2333	0.1860	<u>0.2513</u>
breadcrumbs	0.0854	0.1417	0.1460	0.0495	0.2913	0.1721	0.0766	0.0676	0.0685	0.0802	0.0948	0.0603	0.1225	0.1273	0.1021
model_stock	0.2420	0.1824	0.1975	0.6216	0.4311	0.5721	0.1225	0.1274	0.1185	0.2742	0.1578	0.2012	0.1791	0.1321	0.1578
from	0.2989	0.2306	0.2052	<u>0.6713</u>	0.4369	0.6580	0.2350	0.1560	0.1520	0.2207	0.2087	0.2476	0.2309	0.1506	0.2063
iso_c	0.3278	0.2748	0.3023	0.5049	0.4422	0.6416	0.2331	0.1629	0.1383	0.2736	0.2118	0.2453	0.2615	0.1698	0.1554
free_merge	0.3345	0.2467	0.2822	0.6291	0.4700	0.6246	0.2261	0.1771	0.1396	0.2863	0.1793	0.2475	0.2435	0.1581	0.1545
robustmerge	0.3401	0.2430	0.2256	0.2956	0.2085	0.3942	0.2458	0.1764	0.1653	0.2680	0.2227	0.2301	0.2189	<u>0.2231</u>	0.1707
LAMP-Merge	0.8002	0.7999	0.8007	0.6745	0.6761	0.6763	0.6074	0.6069	0.6076	0.5531	0.5524	0.5517	0.4762	0.4762	0.4762

Table 3: Client-average test accuracy on convnext. Each entry averages the three Dirichlet settings for a fixed client number.

Method	Blood			Derma			Organ-C			Organ-S			Ultrasound		
	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1565	0.1122	0.1384	0.2966	<u>0.4830</u>	0.6688	0.1037	0.1218	0.0944	0.1308	0.1035	0.1416	0.1506	0.1506	0.1252
ties	0.1715	0.1214	0.1524	0.2966	0.1797	0.2713	0.1217	0.0786	0.0716	0.1123	0.0761	0.0796	0.1518	0.1456	0.1291
dare_linear	0.1756	0.1122	0.1345	0.2705	<u>0.4830</u>	0.4830	0.1035	0.0699	0.1085	0.1282	0.1035	0.0797	0.1575	0.1575	0.1264
dare_ties	0.1096	0.1519	0.1497	0.0908	0.2798	0.2451	0.1305	0.1346	0.0770	0.1096	0.0946	0.0709	0.1456	0.1297	0.1500
regmean	0.1565	0.0817	<u>0.1856</u>	0.2633	0.0773	0.2720	0.1143	0.1017	0.0768	0.1014	0.1259	0.1038	0.1575	0.1539	0.1500
fisher	0.1565	0.0999	0.1189	<u>0.4825</u>	0.2638	0.2971	0.1003	0.0974	0.0915	0.0604	0.0517	0.0843	<u>0.1695</u>	0.1360	0.1456
breadcrumbs	0.1516	0.1366	0.1163	0.0524	0.2766	0.4630	0.1031	<u>0.1444</u>	0.1574	0.1516	<u>0.1554</u>	<u>0.2077</u>	0.1384	0.1363	0.1363
model_stock	0.1756	0.1144	0.1778	0.2966	<u>0.4830</u>	0.6688	0.1276	0.1218	0.1385	0.1308	0.1035	0.1935	0.1575	0.1291	0.1479
from	0.1565	0.1083	0.1671	0.2966	0.3406	0.4569	0.1232	0.1224	0.0650	0.1324	0.1105	0.0897	0.1575	<u>0.1578</u>	0.1087
iso_c	<u>0.1947</u>	<u>0.1865</u>	0.1480	0.2971	<u>0.4830</u>	0.6688	0.0991	0.0699	0.0945	0.0789	0.1477	0.0867	0.1518	0.1291	0.1605
free_merge	0.1575	0.1190	0.1384	0.2971	<u>0.4830</u>	0.6688	0.0906	0.1218	0.1385	0.1308	0.1035	0.1416	0.1506	0.1291	<u>0.1695</u>
robustmerge	0.1410	0.1405	0.0752	0.0908	0.4507	0.4830	<u>0.1394</u>	0.1218	<u>0.1793</u>	<u>0.2354</u>	0.0961	<u>0.2077</u>	0.1300	0.1135	0.1363
LAMP-Merge	0.8482	0.8483	0.8493	0.5992	0.5920	<u>0.5852</u>	0.6482	0.6480	0.6495	0.5977	0.5985	0.5990	0.4259	0.4259	0.4259

Table 9: Internal ablation of long-tail prevalence calibration.

Setting	Variable setting	Mean Acc
LAMP-Merge	$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}$	0.6209
No prevalence calibration	$b_c \leftarrow 0$	0.5879
Uniform prevalence prior	$\pi_c \leftarrow 1/C$; C is the number of diagnosis classes.	0.5879
Smoothed prevalence prior	$\pi_c \leftarrow \frac{\sum_i (m_{i,c}+1)}{\sum_k \sum_i (m_{i,k}+1)}$	0.6209

Table 9 shows that removing prevalence calibration or replacing the real prior with a uniform prior reduces mean ACC, confirming that long-tailed prevalence is part of the full method. The smoothed prior remains close to the official method, indicating that LAMP-Merge does not depend on accidental fluctuations of exact counts; bounded calibration remains effective as long as the prior preserves the real long-tailed structure.

D. Hyperparameter Analysis Experiment

Diagnostic prototype reconstruction has one continuous hyperparameter, the prototype head scale s . Long-tail prevalence calibration has one continuous hyperparameter, the calibration strength λ . Each value in figure 3 is evaluated over 180 raw cells and 60 client-average cells, covering five medical datasets, four backbones, three client counts, and three Dirichlet non-IID partitioning parameters.

The results show that LAMP-Merge maintains stable performance across a broad range of hyperparameter values. Varying the prototype-head scaling factor s and long-tail calibration strength λ produces only minor fluctuations, and the default configuration ($s = 20$, $\lambda = 5$) lies within a stable high-performing region. Therefore, our gains primarily stem from diagnostic prototype reconstruction and long-tail prevalence calibration themselves, rather than from finely tuned hyperparameter search.

Table 4: Client-average test accuracy on vit_t. Each entry averages the three Dirichlet settings for a fixed client number.

Method	Blood			Derma			Organ-C			Organ-S			Ultrasound		
	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1343	0.1082	0.1160	0.4638	0.4369	0.3696	0.1657	0.1043	0.0952	0.0844	0.0852	0.1710	0.1450	0.1327	<u>0.1884</u>
ties	0.0796	0.1421	0.1285	0.4780	0.3498	0.4615	0.1341	0.1223	0.0953	0.1192	0.1002	0.1597	<u>0.2001</u>	0.1306	0.1018
dare_linear	<u>0.2397</u>	0.1129	0.1387	0.2850	0.1002	0.1204	0.1534	0.0946	0.0704	0.0748	0.0549	0.1611	0.1366	0.1051	0.1435
dare_ties	0.0735	0.1189	0.1146	0.2735	0.4502	0.3528	0.1252	0.0903	0.1033	0.0541	0.0874	0.0591	0.1447	0.1120	0.1042
regmean	0.1748	0.1417	0.1290	0.5114	0.3667	<u>0.6658</u>	0.1250	0.0964	0.0983	<u>0.1263</u>	0.0717	0.1068	0.1491	0.1384	0.1144
fisher	0.1166	<u>0.1512</u>	<u>0.1944</u>	0.4780	0.2042	0.4820	0.0799	0.1072	0.1424	0.0856	<u>0.1218</u>	0.1222	0.1611	0.1315	0.1420
breadcrumbs	0.1363	0.1039	0.1300	0.0489	0.2572	0.0607	0.0773	0.1000	<u>0.1644</u>	0.1237	0.0483	0.1219	0.1360	0.1141	0.1524
model_stock	0.1940	0.1120	0.1134	0.3805	0.3079	0.2735	0.1488	0.1227	0.1215	0.0893	0.0859	<u>0.1867</u>	0.1599	0.1809	0.1641
from	0.1802	0.0781	0.1405	0.4695	<u>0.5121</u>	0.6688	<u>0.2148</u>	<u>0.1301</u>	0.0972	0.0772	0.0798	0.1485	0.1836	0.1743	0.1776
iso_c	0.1234	0.1483	0.1018	<u>0.5709</u>	0.4529	0.2306	0.1084	0.0567	0.0897	0.0611	0.0531	0.1081	0.1417	0.1527	0.1869
free_merge	0.1989	0.1457	0.1392	0.4630	0.4770	0.3855	0.1425	0.0980	0.0759	0.0727	0.0914	0.1643	0.1405	0.1593	0.1647
robustmerge	0.1482	0.1070	0.1425	0.0347	0.2768	0.1724	0.0954	0.0617	0.1438	0.1260	0.0707	0.1217	0.1698	<u>0.1854</u>	0.1563
LAMP-Merge	0.8523	0.8522	0.8516	0.5970	0.5938	0.5860	0.5805	0.5790	0.5790	0.5406	0.5372	0.5388	0.4645	0.4645	0.4645

Table 5: Client-average test accuracy on swin_tiny. Each entry averages the three Dirichlet settings for a fixed client number.

Method	Blood			Derma			Organ-C			Organ-S			Ultrasound		
	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1565	0.1186	0.1524	0.4569	<u>0.4830</u>	0.6688	0.1068	0.1294	0.0787	0.0682	0.0604	0.1381	0.1527	0.1294	0.1465
ties	0.0972	0.1190	0.1384	0.0841	0.0780	0.6688	0.1160	0.0702	0.0934	0.0734	0.0607	0.1107	0.1695	0.1303	0.1261
dare_linear	0.1565	0.1161	0.1825	0.4569	0.2771	0.4630	0.1155	0.1272	0.0776	0.0640	0.0720	0.1381	0.1743	0.1536	0.1497
dare_ties	0.1384	0.1162	0.1384	0.2961	0.2638	0.6688	0.1276	0.0774	0.1116	0.1194	0.0434	0.1230	<u>0.1791</u>	0.1351	0.1300
regmean	0.1374	0.1715	0.1578	0.2705	0.4630	0.4569	0.1411	0.0773	0.0811	0.1215	0.0821	0.0926	0.1465	0.1105	0.1779
fisher	<u>0.1633</u>	<u>0.1756</u>	0.1565	<u>0.4582</u>	0.0652	0.4718	0.0814	0.0899	0.0978	0.0651	0.0650	0.1092	0.1746	0.1165	0.1755
breadcrumbs	0.1500	0.1365	0.1537	0.0647	0.2771	0.0715	0.1445	<u>0.1783</u>	0.1660	<u>0.2354</u>	<u>0.2077</u>	<u>0.2111</u>	0.1450	0.1312	0.1653
model_stock	0.1374	0.1440	0.1715	0.4569	0.2771	0.4825	0.1110	0.0771	0.1404	0.0671	0.1106	0.1714	0.1539	0.1273	<u>0.1869</u>
from	0.1565	0.1495	0.1671	0.4569	0.2971	0.6688	0.1174	0.1220	0.1051	0.1133	0.1016	0.1380	0.1539	0.1252	0.1476
iso_c	0.1430	0.1722	0.1700	0.3099	0.2840	0.5174	0.1302	0.1327	0.1469	0.1545	0.1727	0.1880	0.1755	<u>0.1731</u>	0.1791
free_merge	0.1154	0.0861	<u>0.1840</u>	0.4569	<u>0.4830</u>	0.6688	0.0959	0.1377	0.0890	0.0682	0.0497	0.1062	0.1518	0.1575	0.1599
robustmerge	0.1374	0.1409	0.1670	0.2377	0.2771	0.0504	<u>0.1601</u>	0.1442	<u>0.1960</u>	0.1673	0.1151	0.2077	0.1761	0.1315	0.1629
LAMP-Merge	0.7737	0.7753	0.7748	0.6374	0.6251	<u>0.6244</u>	0.6651	0.6632	0.6630	0.6078	0.6076	0.6068	0.4816	0.4816	0.4816

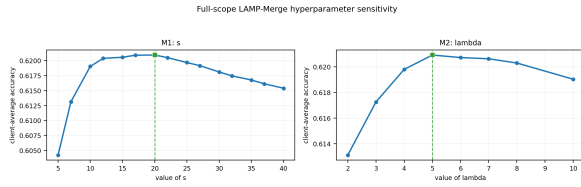


Figure 3: Full-scope hyperparameter sensitivity. The default setting $s = 20$ and $\lambda = 5$ lies in the stable high-accuracy plateau.

E. New-metric Diagnostic Experiment

Although the main experiments use ACC for direct comparison, long-tailed medical distributions can allow single-class or few-class predictors to achieve high ACC. We therefore introduce additional metrics. Let y_n and $\hat{y}_n$ denote the true and predicted labels of the n -th test sample. Let $q_c = N^{-1} \sum_n \mathbf{1}[\hat{y}_n = c]$ be the predicted-class distribution, $p_c = N^{-1} \sum_n \mathbf{1}[y_n = c]$ be the true

class distribution, and $\mathcal{C}_{\text{test}} = \{c \mid \sum_n \mathbf{1}[y_n = c] > 0\}$ be the set of classes observed in the test set. We use the following diagnostic metrics.

- $\text{BA} = |\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \text{Recall}_c$. This is balanced accuracy, or macro recall, and measures whether all observed diagnosis classes are recalled.
- $\text{MacroF1} = |\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \text{F1}_c$. This measures class-balanced precision-recall quality and penalizes methods that only preserve majority-class predictions.
- $\rho = \max_c q_c$. This is the prediction-collapse ratio and measures how strongly predictions concentrate on a single diagnosis class.
- $C_{\text{eff}} = \exp(-\sum_c q_c \log q_c)$. This is the effective number of predicted classes and measures how many diagnosis classes are actually used by the model.
- $\text{TV}(q, p) = \frac{1}{2} \sum_c |q_c - p_c|$. This is the total variation distance between predicted and true class distributions and measures whether the output distribution

Table 6: Client-average test accuracy on CLIP-ViT-B/32. Each entry averages the three Dirichlet settings for a fixed client number.

Method	Blood			Derma			Organ-C			Organ-S			Ultrasound		
	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1857	0.1199	0.1863	0.6688	0.6688	0.5682	0.2360	0.1348	0.1226	0.2978	0.0926	0.1447	0.1602	0.1282	0.1087
ties	0.2472	0.2286	0.1911	0.6677	0.6760	0.4906	0.2868	0.1682	<u>0.1728</u>	0.2797	0.2139	0.1847	0.1668	0.1204	0.1264
dare_linear	0.1715	0.1202	0.1773	0.6688	0.6688	0.5644	0.2203	0.1342	0.1286	<u>0.3028</u>	0.0863	0.1365	0.1518	0.1312	0.1093
dare_ties	0.2588	<u>0.2306</u>	0.1616	0.6717	0.6756	0.4934	0.2498	0.1618	0.1586	0.2796	0.2063	0.1791	0.1752	0.1204	0.1441
regmean	<u>0.2740</u>	0.2043	0.1907	0.6710	0.5520	0.5104	0.2764	<u>0.1924</u>	0.1394	0.3468	<u>0.2295</u>	0.1853	0.1506	0.1423	0.1087
fisher	0.1819	0.1384	0.1604	<u>0.6715</u>	0.4830	0.4828	<u>0.2910</u>	0.1753	0.1463	0.2397	0.0967	0.1403	0.1387	0.1087	0.1288
breadcrumbs	0.1459	0.1202	0.1170	0.6688	0.6657	0.4894	0.1816	0.1236	0.0766	0.2034	0.0954	0.0768	0.1207	0.1330	0.1731
model_stock	0.1689	0.1163	0.1947	0.6688	0.6687	0.5372	0.2121	0.1844	0.0661	0.2550	0.0861	0.1231	0.1539	0.1363	<u>0.1695</u>
from	0.2727	0.1618	0.1942	<u>0.6715</u>	<u>0.6780</u>	<u>0.6399</u>	0.2491	0.1832	0.1943	0.2435	0.2243	<u>0.1886</u>	<u>0.1758</u>	0.1360	0.1264
iso_c	0.1222	0.1303	0.1137	0.6369	0.2118	0.4698	0.1165	0.1050	0.1181	0.0762	0.1004	0.0831	0.1255	<u>0.1479</u>	0.1692
free_merge	0.2145	0.1141	<u>0.2349</u>	0.6688	0.6683	0.5623	0.1501	0.1419	0.0816	0.2489	0.1136	0.0707	0.1488	0.1411	0.1186
robustmerge	0.1680	0.1575	0.1947	0.6688	0.6680	0.4921	0.2487	0.1256	0.1200	0.2701	0.0677	0.1556	0.1575	0.1369	0.1539
LAMP-Merge	0.3908	0.2679	0.2483	0.5646	0.6846	0.6688	0.4460	0.2242	0.1468	0.2670	0.2296	0.2558	0.1947	0.1608	0.1602

Table 7: Dataset-level client-average accuracy in the module ablation. Each dataset contains 15 cells from five model families and three client counts; each cell is averaged over three Dirichlet skew levels.

Dataset	Cells	LAMP-Merge	Prototype only	avg+calibration	avg
Blood	15	0.7156	0.7155	0.1726	0.1699
Ultrasound	15	0.4040	0.4040	0.1551	0.1516
Derma	15	0.6257	0.4975	0.4913	0.5304
Organ-C	15	0.5543	0.5543	0.1331	0.1358
Organ-S	15	0.5096	0.5096	0.1469	0.1488

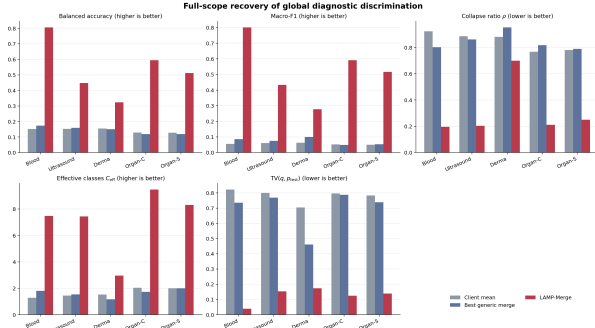


Figure 4: Recovery of global diagnostic discrimination under different datasets.

matches the test-set prevalence.

Table 10 compares LAMP-Merge with the best-performing baseline for each metric. LAMP-Merge consistently and substantially outperforms the existing methods. This indicates that its accuracy gains do not arise from more severe single-class prediction collapse, but rather from more complete global diagnostic discrimination.

F. Visualization Experiment

To visually show the prediction-distribution changes measured by the new diagnostic metrics, we further provide a collapse-recovery visualization in figure 4 and a

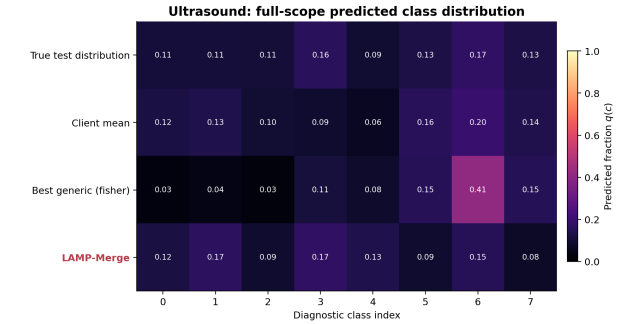


Figure 5: Prediction Distribution on the Ultrasound Dataset

representative predicted-class distribution in figure 5.

G. t-SNE Visualization Experiment

All prototypes live in the shared reference feature space $z = \phi_0(T(x))$. We use t-SNE only for analysis, not for server-side selection or fusion. The joint t-SNE view for Blood and the corresponding LAMP-Merge prototype view are shown in Fig. 6.

H. In-depth Analysis Experiment

1) analysis of the geometry Finally, we conduct an in-depth analysis of the geometry induced by diagnostic prototype reconstruction. This experiment corresponds to the prototype-replacement settings in the internal

Table 8: Internal ablation of diagnostic prototype reconstruction. All settings cover 180 raw cells and 60 client-average cells.

Setting	Variable setting	Mean Acc
LAMP-Merge	$p_c = \sum_i \alpha_{i,c} \mu_{i,c}, e_{i,c} = (n_{i,c} + 1)^\gamma \mathbf{1}[n_{i,c} > 0]$.	0.6209
Classifier-head aggregation	$\mu_{i,c} \leftarrow h_{i,c}; h_{i,c}$ is the class- c local classifier-head direction of client i .	0.2579
Global-feature mean	$\mu_{i,c} \leftarrow g, \forall c; g$ is the class-agnostic global reference-feature mean.	0.2645
Support-only synthetic head	$p_c \leftarrow u_c, \ u_c\ _2 = 1; u_c$ is a fixed random unit direction for class c .	0.1137
Shuffled-label prototype	$p_c \leftarrow p_{\sigma(c)}; \sigma$ is a random permutation over diagnosis labels.	0.2002
Uniform client weight	$\alpha_{i,c} \leftarrow \frac{\mathbf{1}[n_{i,c} > 0]}{\sum_j \mathbf{1}[n_{j,c} > 0]}$.	0.5839
Binary support only	$n_{i,c}, m_{i,c} \leftarrow \mathbf{1}[n_{i,c} > 0]$.	0.5517
Global client-size weight	$e_{i,c} \leftarrow (N_i + 1)^\gamma \mathbf{1}[n_{i,c} > 0], N_i = \sum_k n_{i,k}; N_i$ is the total support of client i .	0.5944

Table 10: Prediction-collapse diagnostics averaged over five medical datasets and four backbones per dataset.

Metric	LAMP-Merge	Best reference	Difference
Accuracy $\uparrow$	0.5885	0.3156	+0.2729
BA / macro recall $\uparrow$	0.5747	0.1810	+0.3937
Macro F1 $\uparrow$	0.5352	0.1123	+0.4229
Collapse ratio $\downarrow$	0.2294	0.7477	-0.5184
Effective classes $\uparrow$	8.0390	2.1360	+5.9030
Pred-true TV $\downarrow$	0.1510	0.6168	-0.4658

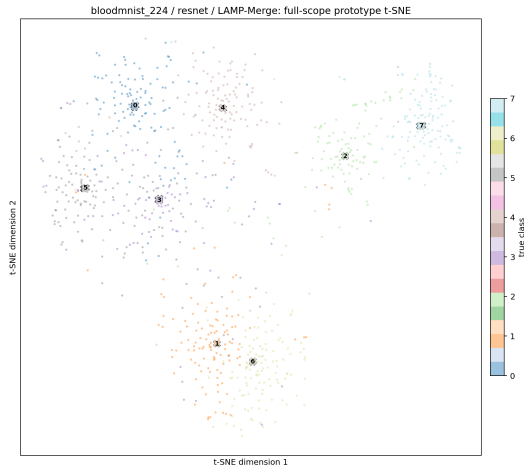


Figure 6: t-SNE Visualization of ResNet Feature Representations on the Blood Dataset

ablation and explains why reference-space class prototypes and class-level support counts jointly provide stable gains. Let $d(a, b) = 1 - \cos(a, b)$ denote cosine distance, and let $\mathcal{I}_c = \{i \mid n_{i,c} > 0\}$ denote the set of clients that contain diagnosis class c . The geometric quantities are defined as follows.

- $D_{\text{pair}} = \frac{2}{C(C-1)} \sum_{c < d} d(p_c, p_d)$. This is the mean distance between global prototypes of different diagnosis classes, measuring whether the reconstructed head forms separable class-level diagnostic directions.
- $D_{\text{nn}} = \frac{1}{C} \sum_c \min_{d \neq c} d(p_c, p_d)$. This is the mean nearest-class distance, measuring the most vulnera-

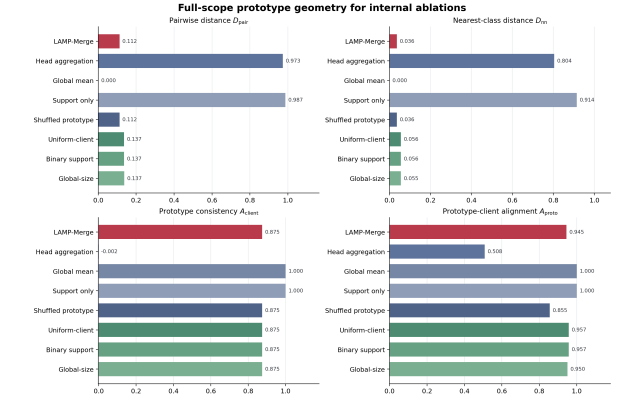


Figure 7: Prototype-geometry comparison on the full benchmark. The figure summarizes pairwise distance, nearest-class distance, prototype consistency, and prototype-client alignment.

ble decision margin around each diagnosis prototype.

- $A_{\text{client}} = \frac{1}{C} \sum_c \text{Avg}_{i < j, i, j \in \mathcal{I}_c} \cos(\mu_{i,c}, \mu_{j,c})$. This is the within-class consistency among client prototypes, measuring whether clients that observe the same diagnosis produce compatible evidence in the shared reference space.
- $A_{\text{proto}} = \frac{1}{C} \sum_c \sum_{i \in \mathcal{I}_c} \alpha_{i,c} \cos(p_c, \mu_{i,c})$. This is the alignment between the server-side global prototype and the contributing client prototypes, measuring whether aggregation preserves client-side class evidence.

Additional Theoretical Analysis Let μ_c^* denote the true class mean in the shared reference space. If client class features have bounded second moments, the prototype estimation error satisfies

$$\mathbb{E} \|p_c - \mu_c^*\|_2^2 \leq \sum_i \alpha_{i,c}^2 \frac{\sigma_c^2}{n_{i,c}} + \text{Bias}_c^2. \quad (11)$$

This upper bound shows that support counts directly control the variance of class-prototype estimation. A client that does not observe class c should not contribute evidence for that class, and a client with more samples should contribute more, but only sublinearly.

Let the true margin between classes c and d for sample x be

$$\Delta_{c,d}(x) = (\mu_c^*)^\top z - (\mu_d^*)^\top z. \quad (12)$$

When the prototype error satisfies

$$\|p_c - \mu_c^*\|_2 + \|p_d - \mu_d^*\|_2 < \frac{\Delta_{c,d}(x)}{\|z\|_2}, \quad (13)$$

the estimated prototypes preserve the class order between c and d .

The role of long-tail prevalence calibration is to introduce long-tail priors in a bounded manner. For prevalence counts $m_{i,c}$, the class prior is

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \quad (14)$$

and the centered log-prior bias is

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_k \log \pi_k \right). \quad (15)$$

As long as λ is bounded, long-tail prevalence calibration does not replace the discriminative directions reconstructed by diagnostic prototype reconstruction; it only calibrates the output toward the real class prevalence. This explains why using long-tail prevalence calibration alone is insufficient, while full LAMP-Merge remains both non-collapsed and prevalence-aware.

Conclusion

We studied one-shot asynchronous medical model merging, where independently trained hospital models must be fused without federated optimization rounds and without exposing raw medical images to the server. Our empirical analysis shows that generic post-hoc merging methods suffer from aggregation-induced predictive collapse under incomplete local diagnostic coverage. We further show that, on long-tailed medical datasets, such collapse can be masked by raw accuracy. We therefore introduce the Long-tail Prevalence Calibration module and perform secondary evaluation using macro recall, macro-F1, and predictive-distribution diagnostics.

LAMP-Merge addresses these two properties through a concise two-module design. Future work will extend its applicability in two directions: first, from medical image classification to segmentation, detection, and multimodal diagnostic tasks; and second, toward stronger privacy-preserving mechanisms, such as differential privacy and secure aggregation, to further reduce potential privacy risks associated with class-level statistics. We will also explore deployment and validation in larger-scale real-world multicenter healthcare settings.